Secure wiring and finish the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Document permanent connections and evaluate added mass or friction.

SUCCESS CRITERIA - I can...

1. Check circuit continuity and mechanical security.
2. Preserve moving-part clearance during decoration.
3. Relate a design revision to an energy transfer or force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “strain relief” and “insulation” to explain your model or result. Sentence starters are optional.

strain relief: support that keeps a pull from stressing an electrical joint

insulation: material that limits unwanted electric conduction

short circuit: an unintended low-resistance path bypassing a component

01 Bring back an earlier idea

At unchanged wheel rpm, circumference becomes 4 times as large.
Compare ideal travel speed and state one real-world limitation.

Word help: circumference = distance around the edge of a circle

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: I = 0 with power disconnected.
Disconnect before soldering/securing wires. Use insulation and strain relief; route wires clear of gears. Follow supervised tool procedures.

Worked reasoning: A secure circuit also needs strain relief: a pull on a wire should not be carried solely by a solder joint. Insulation prevents exposed conductors from making unintended contacts.

Your turn: Draw one wire route showing a connection, insulation, strain relief and clearance from a rotating axle.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Draw your final wiring route.
Mark soldered joints, insulation, strain relief and the switch location.
Identify one possible wheel or gear interference point.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

04 Build the solution

A supplied car mass rises from 200 g to 250 g after decoration.
Find the added mass. For the same forward force, predict whether acceleration increases or decreases.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A wire rubs a wheel.
Explain the possible effect on energy transfer and propose a repair before another powered test.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

06 Choose a method

At fixed current, move from r to 4 r from a long straight wire. Using B proportional to 1/r, compare field strength.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

Record one quality check your partner completed and one revision supported by that check.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Create a power-off inspection checklist with three specific observations that would make you stop before powering the car.
Explain the concern behind each.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3